Name: Surabhi P
Id: COMETFWC032
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https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/platformio

Abstract

The objective of this manual is to show how to control a seven segment display through the Arduino IDE.

1 Components

Component	Value	Quantity
Breadboard	–	1
Resistor	$\geq 220\Omega$	1
Arduino	Uno	1
Seven Segment Display	Common Anode	1
Jumper Wires	–	20

Table 1: Required Components

1.1 Breadboard

The breadboard can be divided into 5 segments. In each green segment, the pins are internally connected to have the same voltage. Similarly, in the central segments, the pins in each column are internally connected.

1.2 Seven Segment Display

The display has eight pins: a, b, c, d, e, f, g, and dot. These take an active LOW input, meaning the LED glows only when connected to ground.

1.3 Arduino

The Arduino Uno has ground pins, analog input pins A0–A3, and digital pins D1–D13. It also has 3.3V and 5V power pins. We use GND, 5V, and digital pins in this manual.

2 Display Control through Hardware

2.1 Powering the Display

Problem 2.1: Plug the display into the breadboard and make the following connections:

Arduino	**Breadboard**
5V	Top Green
GND	Bottom Green

Table 2: Power Connections

Problem 2.2: Make the following connections:

Breadboard	**Display**
5V	Resistor
COM	GND
DOT	–

Table 3: Display Connections

Problem 2.3: Connect the Arduino to the computer. The DOT LED should glow.

2.2 Controlling the Display

Problem 2.4: Generate the number 1 by connecting pins b and c to GND.

Problem 2.5: Repeat to generate the number 2.

Digit	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1

Table 4: Segment Mapping for Digits

3 Display Control through Software

Problem 3.1: Make connections according to the previous table.

Problem 3.2: Download and execute the following code using Arduino IDE:

```
wget https://raw.githubusercontent.com/gadepall/arduino/master/sevenseg/codes/sevenseg/s
```

Problem 3.3: Modify the program to generate digits 0–9.

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